Spring 2010 / EC571

Instructor:Ajay Joshi

Exam 2

Date: April 14, 2010

Name : _____

BU ID : _____

Question	Maximum points	Points scored
1	25	
2	25	
3	25	
4	25	
Total	100	

1. The exam consists of four questions
2. The duration of the exam is 2 hours
3. The maximum grade is 100 points
4. Please write your name on all pages
5. The solutions should be written in the spaces provided on the test sheet
6. If you need to use extra sheets of paper, write your name and BU ID on it, and clearly indicate the question number
7. This exam is open book-open notes
8. Material sharing is not allowed
9. Calculators are allowed
10. Boston University Academic Code of Conduct applies

Question 1

1. Design the following logic function using dynamic logic design technique: $\overline{(AB+C)D}$.

2. Draw the equivalent RC network for circuit drawn in 1.1 and determine the relative widths of the transistors to match the precharge and evaluate cycle times. Assume $R_{sq(PMOS)} = 2R_{sq(NMOS)}$ and a load capacitance of C_L.

3. Calculate the channel widths such that the high-to-low and low-to-high transistion delay = 1 ns. Assume the following technology parameters: $R_{sq(NMOS)}$ = 13 kohm/sq, $R_{sq(PMOS)}$ = 26 kohm/sq, $C_{gNMOS} = C_{gPMOS}$ = 1.5 fF/μm, $C_{dNMOS} = C_{dPMOS}$ = 2 fF/μm, C_L = 50 fF, and the minimum value for channel length and width of NMOS/PMOS is 300 nm and 450 nm, respectively.

Question 2

1. Design the following function using pass-transistor logic design technique: $\overline{EF+G}$. Assume a load capacitance of C_L

2. Explain why this circuit has non-zero static power dissipation

3. Use one transistor to fix the static power dissipation problem. Explain the functioning of the circuit and choice of transistor size.

4. Draw the RC equivalent circuit when E=1, F=1, G=0 and E=0, F=1, G =0 You can ignore keeper transistor. Clearly indicate the R & C contributions of the various transistors.

Question 3

Consider the circuit shown below. Assume a unit delay each for the register and logic block. The registers are positive-edge triggered and have a setup time of one unit. The multiplexer has 2 units of delay.

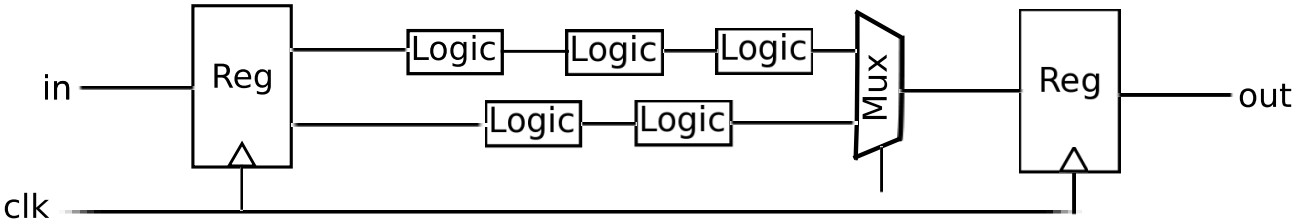


Figure 1

1. Determine the minimum clock period for the above circuit. You can ignore clock skew

2. Determine the minimum clock period considering a positive clock skew of 2 time units

3. Assuming a hold time of 1 time unit, what is the maximum negative clock skew that can be tolerated by this circuit?

Question 4

Design a comparator for 4-bit unsigned binary numbers A and B. Here if A ¿ B then the output is logic 1 else output is logic 0. Draw schematic diagram and identify the critical path.